Stakeholder Analysis

AI-Powered Predictive Maintenance

The Purpose:

This analysis identifies the people and organizations that use, influence, or are affected by the AI-powered predictive maintenance system. For each one, it records their needs and concerns, rates their power and interest, and sets out how the team should engage them. It is the input for the functional and non-functional requirements, the user stories and use cases, and the final SRS.

Assumptions:

- The system is analyzed as if it were deployed in a mid-sized manufacturing plant that runs rotating equipment (pumps, motors, and compressors).
- Sensor data (vibration, temperature, pressure) comes from a public dataset that is replayed as a simulated stream. There is no live SCADA connection.
- The system supports decisions only. It never controls machines or writes back to control systems.
- Power and interest scores are team estimates on a 1–5 scale and will be revisited in each phase.

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1. Stakeholder Identification:

Primary Stakeholders (direct users)

- **Internal:** Maintenance Technician, Reliability/Maintenance Engineer, Machine Operator/Shift Supervisor.

Secondary Stakeholders (indirect or occasional users)

- **Internal:** Maintenance Planner, Plant/Operations Manager, Executive Management/Finance, HSE/Safety Officer, ML Engineer (system maintainer), IT/System Administrator, OT/Control-Systems Engineer.

Tertiary / External Stakeholders

- **External:** Equipment OEMs/vendors, AI & cloud service providers (OpenAI, Hugging Face, hosting), regulators & standards bodies, and the plant's end customers.

Project Stakeholders

- **Academic:** Project Development Team, DEPI Instructors & Evaluation Committee.

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2. Stakeholder Register:

ID	Stakeholder	Type / Category	Role in the System
SH-01	Maintenance Technician	Internal / Primary	Receives alerts, checks the cause of an alert (XAI), asks the LLM assistant, and logs the repairs done.
SH-02	Reliability / Maintenance Engineer	Internal / Primary	Analyzes RUL forecasts and SHAP/LIME explanations, runs what-if simulations, sets alert thresholds, and generates reports.

ID	Stakeholder	Type / Category	Role in the System
SH-03	Machine Operator / Shift Supervisor	Internal / Primary	Checks machine health status and acknowledges or escalates alerts.
SH-04	Maintenance Planner	Internal / Secondary	Uses RUL forecasts and reports to schedule downtime and order spare parts.
SH-05	Plant / Operations Manager	Internal/Key decision-maker	Reviews fleet risk, KPIs, the ROI calculator and PDF reports; approves roll-out.
SH-06	Executive Management / Finance	Internal / Sponsor	Funds the system and judges it on the cost savings shown by the ROI calculator.
SH-07	HSE / Safety Officer	Internal / Secondary	Receives critical alerts and reviews the alert history for safety incidents.
SH-08	ML Engineer (Maintainer)	Internal / Technical	Retrains and versions models (MLflow) and monitors model accuracy and data drift.
SH-09	IT / System Administrator	Internal / Technical	Hosts the app and API, manages users, roles and API keys, and ensures security.
SH-10	OT / Control-Systems Engineer	Internal / Data owner	Owns the sensors and SCADA data; provides read-only data access.
SH-11	Equipment OEMs / Vendors	External / Tertiary	Provide machine operating limits and failure modes; may receive reports for warranty claims.
SH-12	AI & Cloud Service Providers	External / Supplier	Provide the LLM API and hosting; set cost, rate limits and data-use terms.
SH-13	Regulators & Standards Bodies	External / Constraint	ISO 55000 (asset management), ISO 17359 (condition monitoring), Egypt Data Protection Law 151/2020 and occupational-safety rules.
SH-14	Project Development Team	Project / Builder	Designs, builds, tests and documents the system.
SH-15	DEPI Instructors & Evaluators	Academic / Key decision-maker	Evaluate the SRS, diagrams, code, demo and presentation.
SH-16	Plant's End Customers	External / Beneficiary	Do not use the system; benefit from fewer delivery delays.

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3. Stakeholder Needs & Concerns:

Stakeholder	Needs / Expectations	Concerns
SH-01 Technician	Early alerts that show which machine, how urgent and the likely cause; quick answers from the assistant; simple, tablet-friendly screens.	Too many false alarms; complicated dashboards; fear of being replaced or judged by AI.
SH-02 Reliability Engineer	Accurate RUL with a confidence range; an explanation for each prediction; What-if simulation; configurable thresholds; exportable reports.	Black-box predictions they cannot defend; being blamed for a missed failure.
SH-03 Operator	A clear Normal / Warning / Critical status; one-click acknowledge or escalate.	Extra workload; complex interface.
SH-04 Planner	Enough lead time to plan downtime and parts; a list of machines due for maintenance.	Forecasts that change too often to plan around.
SH-05 Plant Manager	Fleet-wide risk overview; downtime and cost KPIs; monthly PDF report; proof of ROI.	Disruption during roll-out; low staff adoption.
SH-06 Executive / Finance	Clear savings figures compared with reactive maintenance.	ROI not proven; ongoing LLM and cloud costs.
SH-07 HSE Officer	Immediate escalation of critical alerts; a full alert history for audits.	Missed dangerous failures; unsafe advice from the assistant.
SH-08 ML Engineer	Experiment tracking, model versioning, drift monitoring, and easy retraining.	Poor data quality; very few real failure examples.

Stakeholder	Needs / Expectations	Concerns
SH-09 IT Admin	Role-based access, secure storage of API keys, logging, backups and simple deployment.	Plant data sent to external LLM services; new security risks.
SH-10 OT Engineer	Read-only data access with no impact on control systems.	Any write-back to PLC/SCADA; load on the plant network.
SH-11 / 12 / 13 External	OEM limits and warranty evidence; fair use of provider APIs; compliance with safety and data-protection rules.	Misuse of machine data; breaches of provider terms; non-compliance.
SH-14 Development Team	Clear, stable and prioritised requirements.	Scope creep; unclear ownership of tasks.
SH-15 DEPI Evaluators	Complete, consistent documents (SRS, UML, code, tests) and a working live demo.	Scope too big to finish; documents that don't match the delivered system.

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4. Power / Interest Analysis:

Each stakeholder was scored from 1 (low) to 5 (high) on power (the ability to approve, block, fund or change the project) and on interest (how much the system affects their work). The grid is split at 3.

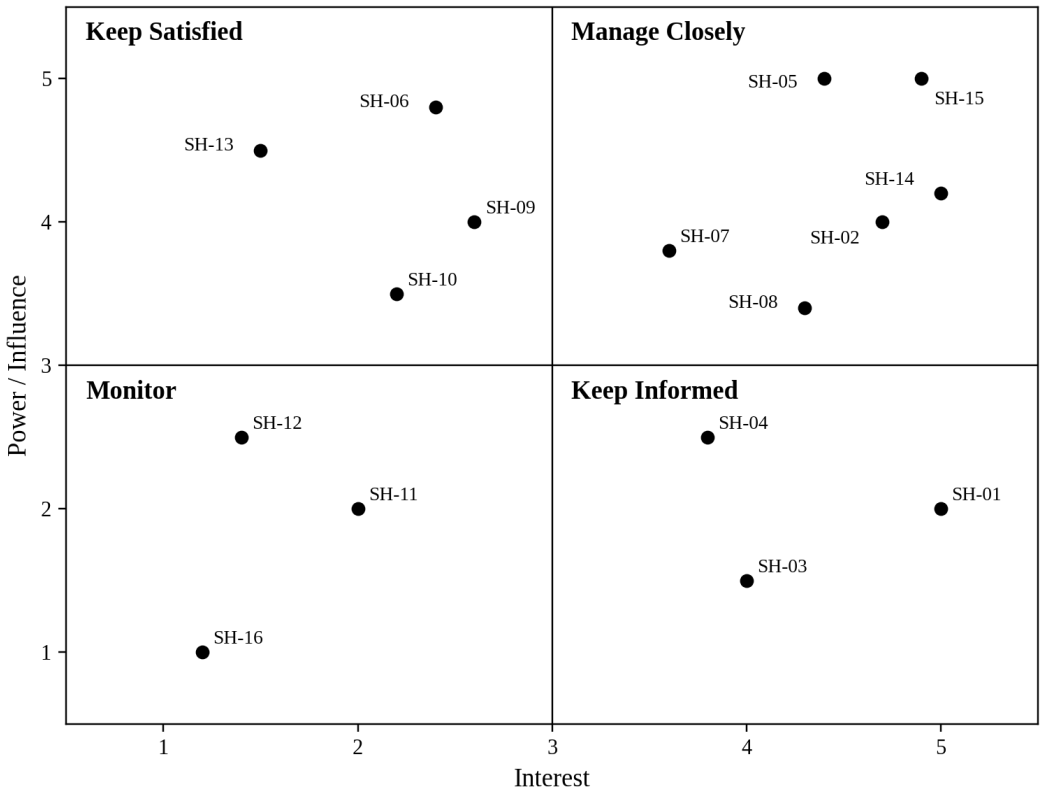


Figure 1: Power / Interest Grid

ID	Stakeholder	Power	Interest	Quadrant
SH-15	DEPI Evaluators	5	5	Manage Closely
SH-05	Plant Manager	5	4.4	Manage Closely
SH-14	Development Team	4.2	5	Manage Closely
SH-02	Reliability Engineer	4	4.7	Manage Closely
SH-07	HSE Officer	3.8	3.6	Manage Closely
SH-08	ML Engineer	3.4	4.3	Manage Closely
SH-06	Executive / Finance	4.8	2.4	Keep Satisfied
SH-13	Regulators & Standards	4.5	1.5	Keep Satisfied
SH-09	IT Admin	4	2.6	Keep Satisfied
SH-10	OT Engineer	3.5	2.2	Keep Satisfied
SH-01	Technician	2	5	Keep Informed
SH-04	Planner	2.5	3.8	Keep Informed
SH-03	Operator	1.5	4	Keep Informed
SH-11	OEMs / Vendors	2	2	Monitor
SH-12	AI & Cloud Providers	2.5	1.4	Monitor
SH-16	End Customers	1	1.2	Monitor

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5. Engagement Strategy:

- **Manage Closely:** Evaluators, Plant Manager, Development Team, Reliability Engineer, HSE Officer, ML Engineer. Involve them in reviews of the SRS, UML, and prototypes; agree on KPIs, alert severity rules, and model metrics with them; and show them demos at each phase.
- **Keep Satisfied:** Finance, Regulators, IT Admin, OT Engineer. Give Finance short ROI summaries. Get IT and OT sign-off on hosting, security, API-key handling, and read-only data access. Build compliance through an audit trail, data protection, and safety alerts.
- **Keep Informed:** Technician, Operator, Planner. Walk them through the prototype and run usability tests on the alert, dashboard, and assistant screens; keep their views simple; and collect their feedback.
- **Monitor:** OEMs, AI & Cloud Providers, End Customers. Use OEM manuals for thresholds, and track provider pricing, rate limits, and data-use terms.

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6. Stakeholder Conflicts & Resolutions:

- **Sensitivity vs. false alarms:** The HSE officer wants every possible failure flagged, while technicians want fewer false alarms. Resolution: severity levels (Info / Warning / Critical), high sensitivity only for the Critical level, and feedback on false alarms.
- **Accuracy vs. explainability:** Deep models (LSTM/GRU) can be more accurate, but engineers and evaluators need explanations. Resolution: apply SHAP/LIME to every model and show the explanation next to every prediction.
- **Cloud LLM vs. data privacy:** Hosted LLMs are powerful, but IT and the regulations restrict plant data leaving the site. Resolution: send only masked or summarized data to the LLM, with a local Hugging Face model as a fallback.

- **Management certainty vs. model uncertainty:** Finance wants firm savings figures, but predictions are probabilistic. Resolution: the ROI calculator shows ranges and assumptions the user can edit.
- **Detailed vs. simple interface:** Engineers want detail; operators and technicians want simplicity. Resolution: a different view for each role.

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7. Key Inputs for the Next Tasks:

- **AI/ML FRs:** Anomaly detection, RUL with a confidence range, severity classification, SHAP/LIME explanations, configurable thresholds, data validation, FastAPI inference, and MLflow tracking and retraining (from SH-01, 02, 07, 08, 10).
- **LLM/UI FRs:** Fleet health dashboard, machine detail and trend views, an alert list with acknowledge/escalate, the What-if simulator, the ROI calculator, the LLM assistant grounded in real logs, PDF reports, and role-based login (from SH-01 to 07 and SH-09).
- **User Stories & Use Cases:** Actors: Technician, Reliability Engineer, Operator, Plant Manager, HSE Officer, ML Engineer, System Admin, Sensor Data Source (system), LLM Service (external system), and Scheduler (time). Base the stories on the needs in Section 3.
- **NFRs:** Performance, accuracy (low false alarms and missed failures), explainability, usability, security, data privacy, availability, maintainability, scalability, auditability, cost control and safety (from Sections 3 and 6).
- **Final SRS:** Sections 1–4 go under “Stakeholders / User Characteristics”; the assumptions go under “Assumptions & Dependencies." Keep the SH-IDs for traceability.